

Module 6 Week A — Stretch: Multilingual NER Comparison

Honors Track. This is an Honors Track stretch assignment. Stretch assignments are for learners who have completed all core assignments, are On Track or Advanced, and are attending consistently. If you are behind on core work, focus there first.

Due: Thursday end of day. Resubmissions are accepted through Saturday of the assignment week.

Prerequisite: None – this stretch builds on base lab and integration concepts from Week A.

The Challenge

The climate articles dataset contains both English and Arabic text. The base assignments only process English texts because the spaCy `en_core_web_sm` model handles English only. But in professional Natural Language Processing (NLP) work in the Middle East and North Africa (MENA) region, bilingual data is the norm.

Compare Named Entity Recognition (NER) performance across English and Arabic text using multilingual models. Load models that can process both languages, run them on both the English and Arabic portions of the climate dataset, and analyze how NER quality differs between languages.

This directly prepares you for building real-world NLP systems in Jordan’s bilingual professional environment, where the same pipeline must handle English reports, Arabic news, and mixed-language documents.

What You’ll Learn

- Working with multilingual NLP models (spaCy `xx_ent_wiki_sm`, Hugging Face `xlm-roberta`)
- Cross-lingual NER evaluation: same entity types, different language challenges
- Arabic text processing considerations: right-to-left display, diacritics, morphological complexity
- Analyzing model failure modes across languages
- Connecting NLP capabilities to MENA professional contexts

Outcome

Your deliverable extends your Lab 6A repo (`m6-16a-{your-username}`) on a new branch — the same convention as the Tuesday stretch — so the climate articles dataset is already in place. The deliverable contains:

Artifact	Description
Multilingual NER script	Runs both a multilingual spaCy model and a multilingual HF model on English and Arabic texts
Comparison table	Entity counts, entity types, and entity density (entities per 100 words) broken down by language and model. Note: the gold standard has no Arabic annotations, so Arabic precision/recall/F1 cannot be measured — Arabic evaluation in this stretch is qualitative (inspect a sample of outputs and report what each model finds vs. misses)
2-paragraph analysis	(a) What entity types are harder in Arabic vs. English, and why. (b) What this means for bilingual NLP applications in the MENA region

Constraints:

- Use at least two multilingual models:
 - spaCy: `xx_ent_wiki_sm` (install: `python -m spacy download xx_ent_wiki_sm`)
 - Hugging Face: `Davlan/xlm-roberta-base-wikiann-ner` or similar multilingual NER model
- Process at least 20 English texts and 20 Arabic texts from the climate dataset
- For each language × model combination, report: total entities found, entity count per type (ORG, GPE, PERSON, DATE, etc.), and 3 example entities with their labels
- The comparison table should make cross-lingual differences immediately visible
- The analysis must reference specific examples from the dataset – not generic observations about Arabic NLP

Tips

- Arabic NER has different challenges than English: proper nouns do not have case distinction (no capital letters), entity boundaries are harder to detect in morphologically rich text, and diacritics (tashkeel) may or may not be present.
- The multilingual models use different entity label schemas than the English-only models — `xx_ent_wiki_sm` emits `PER / LOC / ORG / MISC`, while spaCy’s English models (and the lab’s gold standard) emit `PERSON / GPE / DATE / ORG` / etc. You have two options for the comparison table; pick one and state it explicitly:
 - Option A — keep native labels.** Show each model’s labels as-is in the table. Cleanest when you are not computing cross-model match rates.
 - Option B — apply a label mapping.** Map multilingual labels onto the English schema using a table like this, then use the mapped labels for cross-model comparison:

Multilingual label	Mapped label	Notes
<code>PER</code>	<code>PERSON</code>	Direct equivalent
<code>LOC</code>	<code>GPE</code> (default) or <code>LOC</code>	<code>xx_ent_wiki_sm</code> does not distinguish geopolitical entities; choose one and document
<code>ORG</code>	<code>ORG</code>	Direct equivalent
<code>MISC</code>	(drop or keep as <code>MISC</code>)	No clean English equivalent

- Some Arabic texts in the dataset are short. If a model finds zero entities in a text, that is a data point – not an error. Report the “no entities found” rate per language.
- If model downloads are slow, start the downloads early and work on your analysis framework while they complete.

Submission

- In your Lab 6A repo (`m6-16a-{your-username}`), create a new branch named `stretch-multilingual-ner` off `main`
- Add your multilingual NER script (e.g., `stretch_multilingual_ner.py`), the comparison table, and your written analysis (e.g., `stretch_analysis.md`). The lab’s `data/climate_articles.csv` is already in the repo
- Push the branch and **open a separate pull request from `stretch-multilingual-ner` to `main`** — do not add these commits to your existing Lab 6A PR or to the Tuesday stretch PR. Each stretch is a separate PR
- Paste the **stretch PR URL** into TalentLMS -> Module 6 Week A -> Stretch 6A-S2

